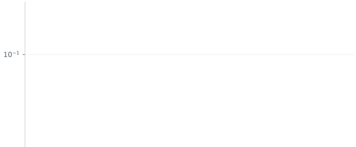
Figure 12b · One epoch, full data

RTX 6000 Ada · 32 test cases per family · Weight EMA

Fast paths and both decoder paths: measured timings. Other native screens: shared timing proxy.

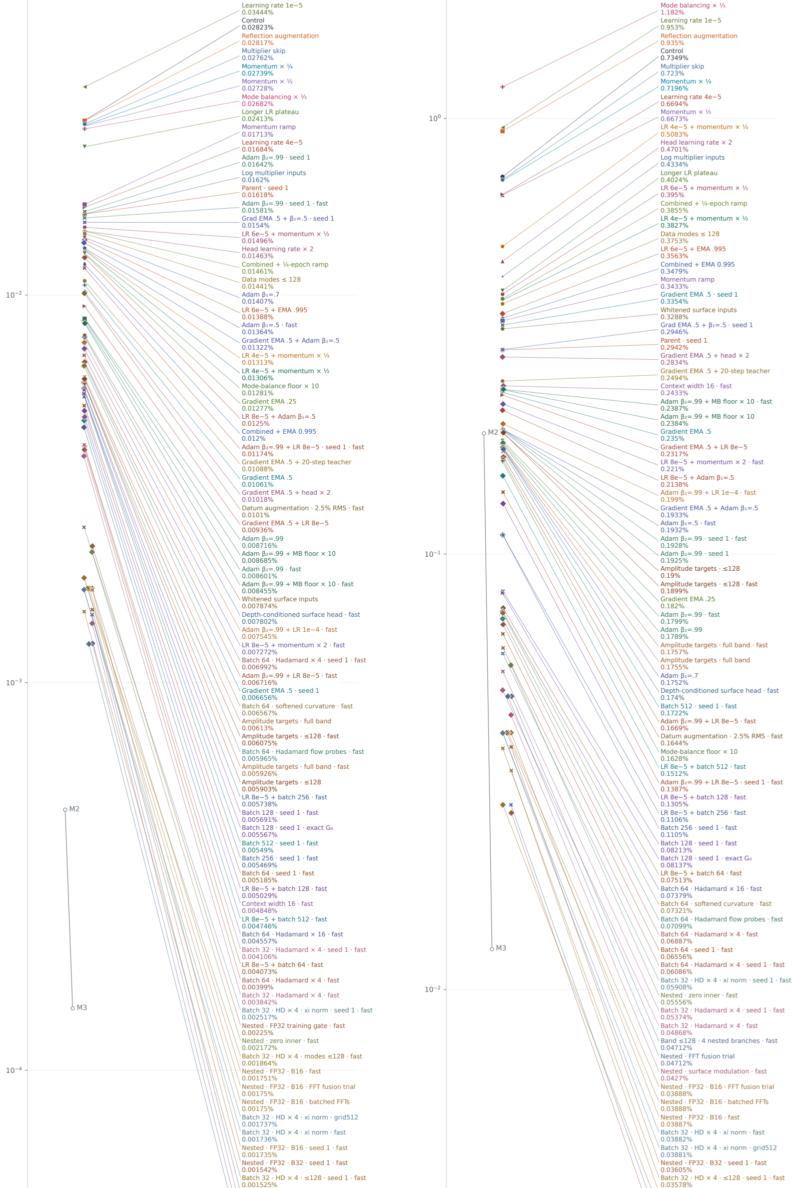
Stokes

T = 20

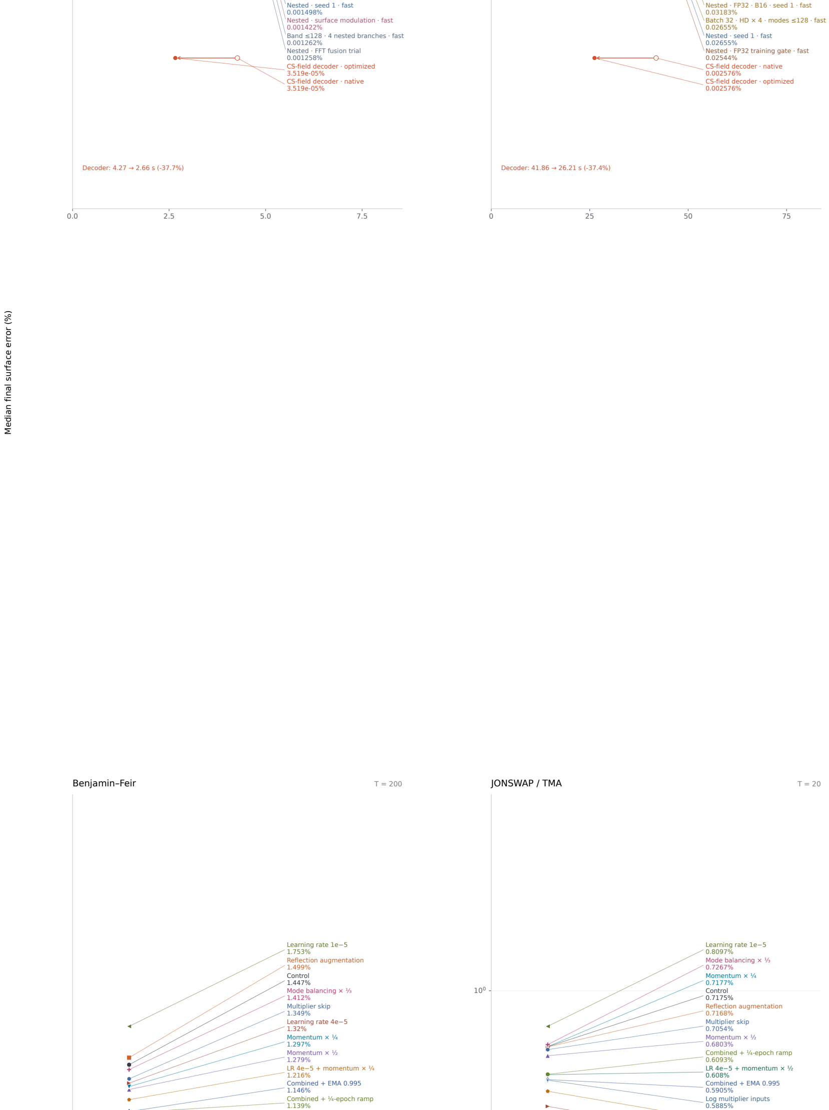


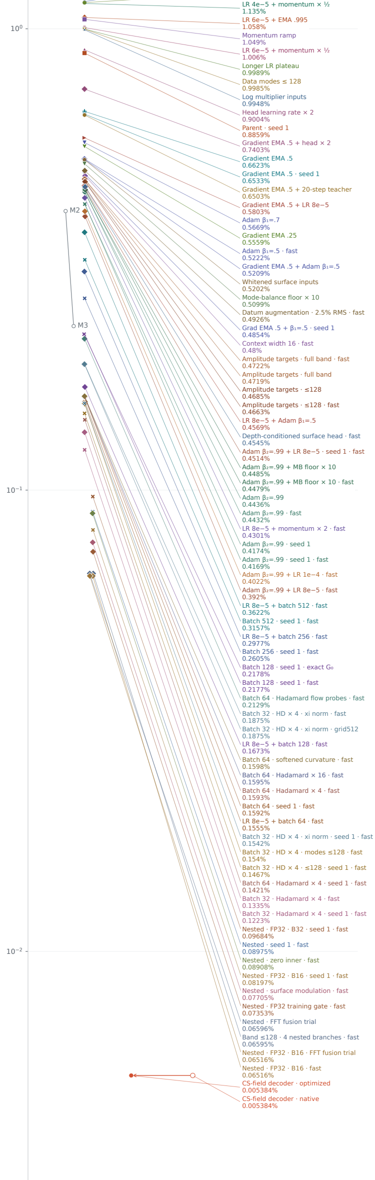
Tanaka

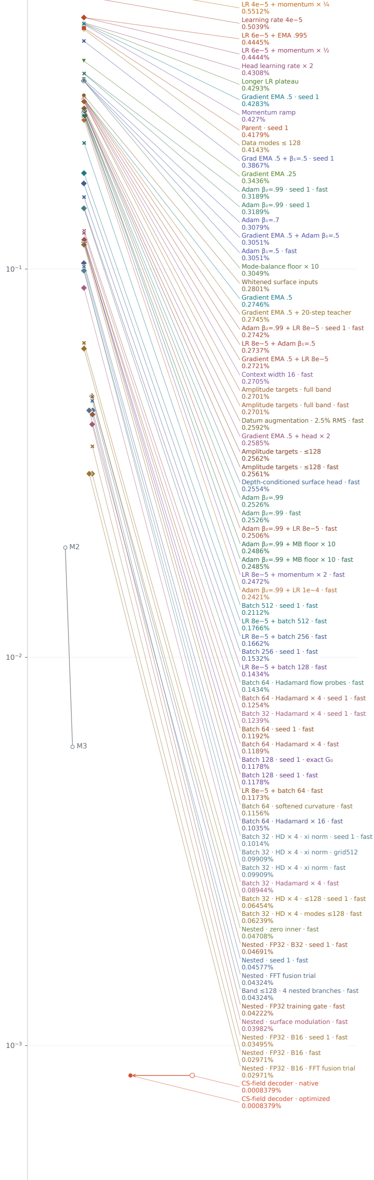
T = 200

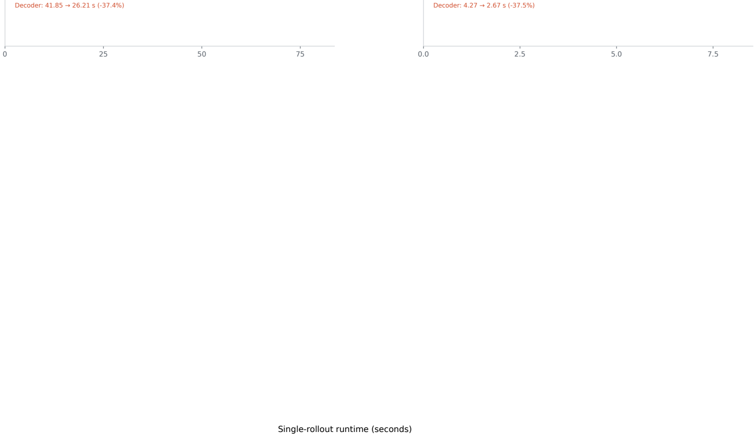


Learning rate 1e−5
0.03444%
Control
0.02823%
Reflection augmentation
0.02817%
Multiplier skip
0.02762%
Momentum × ¼
0.02739%
Momentum × ½
0.02728%
Mode balancing × ⅓
0.02682%
Longer LR plateau
0.02413%
Momentum ramp
0.01713%
Learning rate 4e−5
0.01684%
Adam β₂=.99 · seed 1
0.01642%
Log multiplier inputs
0.0162%
Parent · seed 1
0.01618%
Adam β₂=.99 · seed 1 · fast
0.01581%
Grad EMA .5 + β₁=.5 · seed 1
0.0154%
LR 6e−5 + momentum × ½
0.01496%
Head learning rate × 2
0.01463%
Combined + ¼-epoch ramp
0.01461%
Data modes ≤ 128
0.01441%
Adam β₁=.7
0.01407%
LR 6e−5 + EMA .995
0.01388%
Adam β₁=.5 · fast
0.01364%
Gradient EMA .5 + Adam β₁=.5
0.01322%
LR 4e−5 + momentum × ¼
0.01313%
LR 4e−5 + momentum × ½
0.01306%
Mode-balance floor × 10
0.01281%
Gradient EMA .25
0.01277%
LR 8e−5 + Adam β₁=.5
0.0125%
Combined + EMA 0.995
0.012%
Adam β₂=.99 + LR 8e−5 · seed 1 · fast
0.01174%
Gradient EMA .5 + 20-step teacher
0.01088%
Gradient EMA .5
0.01061%
Gradient EMA .5 + head × 2
0.01018%
Datum augmentation · 2.5% RMS · fast
0.0101%
Gradient EMA .5 + LR 8e−5
0.00936%
Adam β₂=.99
0.008716%
Adam β₂=.99 + MB floor × 10
0.008685%
Adam β₂=.99 · fast
0.008601%
Adam β₂=.99 + MB floor × 10 · fast
0.008455%
Whitened surface inputs
0.007874%
Depth-conditioned surface head · fast
0.007802%
Adam β₂=.99 + LR 1e−4 · fast
0.007545%
LR 8e−5 + momentum × 2 · fast
0.007272%
Batch 64 · Hadamard × 4 · seed 1 · fast
0.006992%
Adam β₂=.99 + LR 8e−5 · fast
0.006716%
Gradient EMA .5 · seed 1
0.006656%
Batch 64 · softened curvature · fast
0.006567%
Amplitude targets · full band
0.00613%
Amplitude targets · ≤128 · fast
0.006075%
Batch 64 · Hadamard flow probes · fast
0.005965%
Amplitude targets · full band · fast
0.005926%
Amplitude targets · ≤128
0.005903%
LR 8e−5 + batch 256 · fast
0.005738%
Batch 128 · seed 1 · fast
0.005691%
Batch 128 · seed 1 · exact G₀
0.005567%
Batch 512 · seed 1 · fast
0.00549%
Batch 256 · seed 1 · fast
0.005469%
Batch 64 · seed 1 · fast
0.005185%
LR 8e−5 + batch 128 · fast
0.005029%
Context width 16 · fast
0.004848%
LR 8e−5 + batch 512 · fast
0.004746%
Batch 64 · Hadamard × 16 · fast
0.004557%
Batch 32 · Hadamard × 4 · seed 1 · fast
0.004106%
LR 8e−5 + batch 64 · fast
0.004073%
Batch 64 · Hadamard × 4 · fast
0.00399%
Batch 32 · Hadamard × 4 · fast
0.003842%
Batch 32 · HD × 4 · xi norm · seed 1 · fast
0.002517%
Nested · FP32 training gate · fast
0.00225%
Nested · zero inner · fast
0.002172%
Batch 32 · HD × 4 · modes ≤128 · fast
0.001864%
Nested · FP32 · B16 · fast
0.001751%
Nested · FP32 · B16 · FFT fusion trial
0.00175%
Nested · FP32 · B16 · batched FFTs
0.00175%
Batch 32 · HD × 4 · xi norm · grid512
0.001737%
Batch 32 · HD × 4 · xi norm · fast
0.001736%
Nested · FP32 · B16 · seed 1 · fast
0.001735%
Nested · FP32 · B32 · seed 1 · fast
0.001542%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast
0.001525%
10^{-2}
10^{-3}
10^{-4}
M2
M3
Mode balancing × ⅓
1.182%
Learning rate 1e−5
0.953%
Reflection augmentation
0.935%
Control
0.7349%
Multiplier skip
0.723%
Momentum × ¼
0.7196%
Learning rate 4e−5
0.6694%
Momentum × ½
0.6673%
LR 4e−5 + momentum × ¼
0.5083%
Head learning rate × 2
0.4701%
Log multiplier inputs
0.4334%
Longer LR plateau
0.4024%
LR 6e−5 + momentum × ½
0.395%
Combined + ¼-epoch ramp
0.3855%
LR 4e−5 + momentum × ½
0.3827%
Data modes ≤ 128
0.3753%
LR 6e−5 + EMA .995
0.3563%
Combined + EMA 0.995
0.3479%
Momentum ramp
0.3433%
Gradient EMA .5 · seed 1
0.3354%
Whitened surface inputs
0.3288%
Grad EMA .5 + β₁=.5 · seed 1
0.2946%
Parent · seed 1
0.2942%
Gradient EMA .5 + head × 2
0.2834%
Gradient EMA .5 + 20-step teacher
0.2494%
Context width 16 · fast
0.2433%
Adam β₂=.99 + MB floor × 10 · fast
0.2387%
Adam β₂=.99 + MB floor × 10
0.2384%
Gradient EMA .5
0.235%
Gradient EMA .5 + LR 8e−5
0.2317%
LR 8e−5 + momentum × 2 · fast
0.221%
LR 8e−5 + Adam β₁=.5
0.2138%
Adam β₂=.99 + LR 1e−4 · fast
0.199%
Gradient EMA .5 + Adam β₁=.5
0.1933%
Adam β₁=.5 · fast
0.1932%
Adam β₂=.99 · seed 1 · fast
0.1928%
Adam β₂=.99 · seed 1
0.1925%
Amplitude targets · ≤128
0.19%
Amplitude targets · ≤128 · fast
0.1899%
Gradient EMA .25
0.182%
Adam β₂=.99 · fast
0.1799%
Adam β₂=.99
0.1789%
Amplitude targets · full band · fast
0.1757%
Amplitude targets · full band
0.1755%
Adam β₁=.7
0.1752%
Depth-conditioned surface head · fast
0.174%
Batch 512 · seed 1 · fast
0.1722%
Adam β₂=.99 + LR 8e−5 · fast
0.1669%
Datum augmentation · 2.5% RMS · fast
0.1644%
Mode-balance floor × 10
0.1628%
LR 8e−5 + batch 512 · fast
0.1512%
Adam β₂=.99 + LR 8e−5 · seed 1 · fast
0.1387%
LR 8e−5 + batch 128 · fast
0.1305%
LR 8e−5 + batch 256 · fast
0.1106%
Batch 256 · seed 1 · fast
0.1105%
Batch 128 · seed 1 · fast
0.08213%
Batch 128 · seed 1 · exact G₀
0.08137%
LR 8e−5 + batch 64 · fast
0.07513%
Batch 64 · Hadamard × 16 · fast
0.07379%
Batch 64 · softened curvature · fast
0.07321%
Batch 64 · Hadamard flow probes · fast
0.07099%
Batch 64 · Hadamard × 4 · fast
0.06887%
Batch 64 · seed 1 · fast
0.06556%
Batch 64 · Hadamard × 4 · seed 1 · fast
0.06086%
Batch 32 · HD × 4 · xi norm · seed 1 · fast
0.05908%
Nested · zero inner · fast
0.05556%
Batch 32 · Hadamard × 4 · seed 1 · fast
0.05374%
Batch 32 · Hadamard × 4 · fast
0.04868%
Band ≤128 · 4 nested branches · fast
0.04712%
Nested · FFT fusion trial
0.04712%
Nested · surface modulation · fast
0.0427%
Nested · FP32 · B16 · FFT fusion trial
0.03888%
Nested · FP32 · B16 · batched FFTs
0.03888%
Nested · FP32 · B16 · fast
0.03887%
Batch 32 · HD × 4 · xi norm · fast
0.03882%
Batch 32 · HD × 4 · xi norm · grid512
0.03881%
Nested · FP32 · B32 · seed 1 · fast
0.03605%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast
0.03578%
10^{0}
10^{-1}
10^{-2}
M2
M3




10^{0}
10^{-1}
10^{-2}
M2
M3
LR 4e−5 + momentum × ½
1.135%
LR 6e−5 + EMA .995
1.058%
Momentum ramp
1.049%
LR 6e−5 + momentum × ½
1.006%
Longer LR plateau
0.9989%
Data modes ≤ 128
0.9985%
Log multiplier inputs
0.9948%
Head learning rate × 2
0.9004%
Parent · seed 1
0.8859%
Gradient EMA .5 + head × 2
0.7403%
Gradient EMA .5
0.6623%
Gradient EMA .5 · seed 1
0.6533%
Gradient EMA .5 + 20-step teacher
0.6503%
Gradient EMA .5 + LR 8e−5
0.5803%
Adam β_1=.7
0.5669%
Gradient EMA .25
0.5559%
Adam β_1=.5 · fast
0.5222%
Gradient EMA .5 + Adam β_1=.5
0.5209%
Whitened surface inputs
0.5202%
Mode-balance floor × 10
0.5099%
Datum augmentation · 2.5% RMS · fast
0.4926%
Grad EMA .5 + β_1=.5 · seed 1
0.4854%
Context width 16 · fast
0.48%
Amplitude targets · full band · fast
0.4722%
Amplitude targets · full band
0.4719%
Amplitude targets · ≤128
0.4685%
Amplitude targets · ≤128 · fast
0.4663%
LR 8e−5 + Adam β_1=.5
0.4569%
Depth-conditioned surface head · fast
0.4545%
Adam β_2=.99 + LR 8e−5 · seed 1 · fast
0.4514%
Adam β_2=.99 + MB floor × 10
0.4485%
Adam β_2=.99 + MB floor × 10 · fast
0.4479%
Adam β_2=.99
0.4436%
Adam β_2=.99 · fast
0.4432%
LR 8e−5 + momentum × 2 · fast
0.4301%
Adam β_2=.99 · seed 1
0.4174%
Adam β_2=.99 · seed 1 · fast
0.4169%
Adam β_2=.99 + LR 1e−4 · fast
0.4022%
Adam β_2=.99 + LR 8e−5 · fast
0.392%
LR 8e−5 + batch 512 · fast
0.3622%
Batch 512 · seed 1 · fast
0.3157%
LR 8e−5 + batch 256 · fast
0.2977%
Batch 256 · seed 1 · fast
0.2605%
Batch 128 · seed 1 · exact G_0
0.2178%
Batch 128 · seed 1 · fast
0.2177%
Batch 64 · Hadamard flow probes · fast
0.2129%
Batch 32 · HD × 4 · xi norm · fast
0.1875%
Batch 32 · HD × 4 · xi norm · grid512
0.1875%
LR 8e−5 + batch 128 · fast
0.1673%
Batch 64 · softened curvature · fast
0.1598%
Batch 64 · Hadamard × 16 · fast
0.1595%
Batch 64 · Hadamard × 4 · fast
0.1593%
Batch 64 · seed 1 · fast
0.1592%
LR 8e−5 + batch 64 · fast
0.1555%
Batch 32 · HD × 4 · xi norm · seed 1 · fast
0.1542%
Batch 32 · HD × 4 · modes ≤128 · fast
0.154%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast
0.1467%
Batch 64 · Hadamard × 4 · seed 1 · fast
0.1421%
Batch 32 · Hadamard × 4 · fast
0.1335%
Batch 32 · Hadamard × 4 · seed 1 · fast
0.1223%
Nested · FP32 · B32 · seed 1 · fast
0.09684%
Nested · seed 1 · fast
0.08975%
Nested · zero inner · fast
0.08908%
Nested · FP32 · B16 · seed 1 · fast
0.08197%
Nested · surface modulation · fast
0.07705%
Nested · FP32 training gate · fast
0.07353%
Nested · FFT fusion trial
0.06596%
Band ≤128 · 4 nested branches · fast
0.06595%
Nested · FP32 · B16 · FFT fusion trial
0.06516%
Nested · FP32 · B16 · fast
0.06516%
CS-field decoder · optimized
0.005384%
CS-field decoder · native
0.005384%


10^{-1}
10^{-2}
10^{-3}
M2
M3
LR 4e−5 + momentum × ¼
0.5512%
Learning rate 4e−5
0.5039%
LR 6e−5 + EMA .995
0.4445%
LR 6e−5 + momentum × ½
0.4444%
Head learning rate × 2
0.4308%
Longer LR plateau
0.4293%
Gradient EMA .5 · seed 1
0.4283%
Momentum ramp
0.427%
Parent · seed 1
0.4179%
Data modes ≤ 128
0.4143%
Grad EMA .5 + β_1=.5 · seed 1
0.3867%
Gradient EMA .25
0.3436%
Adam β_2=.99 · seed 1 · fast
0.3189%
Adam β_2=.99 · seed 1
0.3189%
Adam β_1=.7
0.3079%
Gradient EMA .5 + Adam β_1=.5
0.3051%
Adam β_1=.5 · fast
0.3051%
Mode-balance floor × 10
0.3049%
Whitened surface inputs
0.2801%
Gradient EMA .5
0.2746%
Gradient EMA .5 + 20-step teacher
0.2745%
Adam β_2=.99 + LR 8e−5 · seed 1 · fast
0.2742%
LR 8e−5 + Adam β_1=.5
0.2737%
Gradient EMA .5 + LR 8e−5
0.2721%
Context width 16 · fast
0.2705%
Amplitude targets · full band
0.2701%
Amplitude targets · full band · fast
0.2701%
Datum augmentation · 2.5% RMS · fast
0.2592%
Gradient EMA .5 + head × 2
0.2585%
Amplitude targets · ≤128
0.2562%
Amplitude targets · ≤128 · fast
0.2561%
Depth-conditioned surface head · fast
0.2554%
Adam β_2=.99
0.2526%
Adam β_2=.99 · fast
0.2526%
Adam β_2=.99 + LR 8e−5 · fast
0.2506%
Adam β_2=.99 + MB floor × 10
0.2486%
Adam β_2=.99 + MB floor × 10 · fast
0.2485%
LR 8e−5 + momentum × 2 · fast
0.2472%
Adam β_2=.99 + LR 1e−4 · fast
0.2421%
Batch 512 · seed 1 · fast
0.2112%
LR 8e−5 + batch 512 · fast
0.1766%
LR 8e−5 + batch 256 · fast
0.1662%
Batch 256 · seed 1 · fast
0.1532%
LR 8e−5 + batch 128 · fast
0.1434%
Batch 64 · Hadamard flow probes · fast
0.1434%
Batch 64 · Hadamard × 4 · seed 1 · fast
0.1254%
Batch 32 · Hadamard × 4 · seed 1 · fast
0.1239%
Batch 64 · seed 1 · fast
0.1192%
Batch 64 · Hadamard × 4 · fast
0.1189%
Batch 128 · seed 1 · exact G_0
0.1178%
Batch 128 · seed 1 · fast
0.1178%
LR 8e−5 + batch 64 · fast
0.1173%
Batch 64 · softened curvature · fast
0.1156%
Batch 64 · Hadamard × 16 · fast
0.1035%
Batch 32 · HD × 4 · xi norm · seed 1 · fast
0.1014%
Batch 32 · HD × 4 · xi norm · grid512
0.09909%
Batch 32 · HD × 4 · xi norm · fast
0.09909%
Batch 32 · Hadamard × 4 · fast
0.08944%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast
0.06454%
Batch 32 · HD × 4 · modes ≤128 · fast
0.06239%
Nested · zero inner · fast
0.04708%
Nested · FP32 · B32 · seed 1 · fast
0.04691%
Nested · seed 1 · fast
0.04577%
Nested · FFT fusion trial
0.04324%
Band ≤128 · 4 nested branches · fast
0.04324%
Nested · FP32 training gate · fast
0.04222%
Nested · surface modulation · fast
0.03982%
Nested · FP32 · B16 · seed 1 · fast
0.03495%
Nested · FP32 · B16 · fast
0.02971%
Nested · FP32 · B16 · FFT fusion trial
0.02971%
CS-field decoder · native
0.0008379%
CS-field decoder · optimized
0.0008379%


Decoder: 41.85 → 26.21 s (-37.4%)
0
25
50
75
Decoder: 4.27 → 2.67 s (-37.5%)
0.0
2.5
5.0
7.5
Single-rollout runtime (seconds)